

2026



Analyst Lab Africa

Experience Lab Internship

FinTrust Digital Bank: Develop and Integrate

Data Analytics Track

Intern: Emmanuel Asamoah Oppong

Reporting Date: Sunday, 4th October 2026

Week 3: COMPREHENSIVE ANALYSIS REPORT

1. Project Overview

This Report presents the Week 3 deliverable for the FinTrust Digital Bank project on the Data Analytics track. Building directly on the week 2 exploratory work, Week 3 advanced the project. The Week 2 exploratory notebook and descriptive findings were deepened through advanced SQL analysis, extended Python analytics, and a substantially developed Power BI dashboard, and every major findings was formally validated against the underlying records.

Key findings for Week 2 Analysis: FinTrust processed 12,000 transactions worth NGN 560.4 million across a base of 1,500 customers. The overall transaction success rate stands at 90.5%, meaning roughly one in ten transactions fails, is reversed, or remains pending, a material reliability and customer-experience issue. Transfers are the engine of the business, contributing NGN 218 million (39%) of successful transaction value, yet they are also the single largest source of risk-review flags. International transactions, although only 2.5% of volume, are flagged for risk review at 36.9%, nearly double the domestic rate of 18.9% and high-value transactions above NGN 100,000 are flagged, which is 30.0% (518 of 1,729 transactions). The goal of Week 3 is to convert descriptive Week 2 outputs into a decision-oriented business intelligence solution that is accurate, validated, and ready for final integration and presentation in Week 4.

All customer, transaction and flag data are synthetic and created for educational purposes; the Risk_Review_Flag is a synthetic educational target and is not real fraud.

2. Week 3 Objectives

1. Review Week 2 outputs and identify at least three areas requiring improvement or deeper analysis (Assignment Part A).
2. Expand the SQL analysis with at least eight additional meaningful analyses using JOINS, GROUP BY, CASE statements, subqueries, CTEs, aggregates, and window functions (Part B).
3. Extend the Python analysis with at least five additional analytical outputs and visualisations, including comparative, trend, and segmentation analysis (Part C).
4. Substantially develop the Power BI dashboard with KPI cards, trends, customer, transaction and risk analysis, slicers, and drill-downs (Part D).

5. Formally validate at least five major business findings using a Finding → Evidence → Validation → Business Meaning structure (Part E).
6. Develop at least five evidence-based management recommendations (Part F).
7. Demonstrate at least one meaningful test-and-refine improvement cycle, and leave the solution ready for Week 4 final testing, refinement, documentation, and presentation.

3. Part A: Review of Week 2 Analysis

Week 2 established the data-quality baseline, a Mobile-App-dominant channel mix, Transfers as the leading transaction type by value, and an initial decile-based concentration finding (top 10% of customers hold roughly a quarter of value). It did not, however, go far enough in three specific respects, which became the focus of Week 3 development:

Gap Identified	Evidence from Week 2	Week 3 Improvement
Findings were descriptive, not independently validated	Headline figures were stated once, with no re-computation across tools	Every major figure (totals, success rate, risk rate, concentration finding) was re-computed independently in Excel, Python, and Power BI and reconciled in a single KPI_Check table
No customer-level or behavioural depth	Week 2 analysis stopped at transaction level cuts; there was no way to say which customers mattered most beyond a simple value ranking	A full customer-level model was built (1,500 rows, one per customer), and an RFM (Recency, Frequency, Monetary) segmentation was layered on top, in both Excel and Python
Risk-review flag was noted but not profiled	Week 2 mentioned the flag existed but did not break it down by type, channel, value band, or international status	The flag was profiled across five dimensions: transaction type, channel, the full type×channel matrix, value band, and domestic vs. international status

4. Part B: Advanced SQL Analysis

The SQL layer carries forward from Week 2's queries. Rather than re-litigate that work here, the key outputs that feed directly into this week's validated findings are the channel-level value/reliability breakdown and the type-level value ranking shown in Sections 5 and 6 below, both of which were run as grouped aggregations with CASE-based status classification.

5. Part C: Advanced Excel Analysis

The Excel layer rebuilt the transaction log into a customer-level model and used it to answer questions the week 2 transaction-level view couldn't reach.

- **Customer Model (Cust_Model):** A 1,500 row table, one row per customer, aggregating transaction count, success rate, successful value, flag count, and recency from the 12,000-row log. This the foundation every other Excel sheet builds on, and it was checksummed against the raw data before use: transaction counts sum to exactly 12,000 and flag counts sum to exactly 2,352, confirming the rebuild is complete and accurate.
- **RFM segment (RFM_Summary):** Each customer was scored on Recency, Frequency, and Monetary value and assigned to one of four segments:

Segment	Customer	Successful value	% of value
At Risk	600 (40.0%)	₦177,875,451	34%
Loyal	412 (27.5%)	₦145,686,543	28.5%
Champion	160 (10.7%)	₦105,380,871	20.6%
Regular	328 (21.90%)	₦81,864,577	16.0%

Business Insight: At Risk is both the largest segment by headcount and the largest by value held; this is single is the single biggest finding in the whole workbook. It means FinTrust's largest pool of revenue sits with customers who are transacting less frequently and less recently than the rest of the base. Champions, by contrast, are a small group (10.7% of customers) carrying a disproportionate 20.6% of value, losing even a few of the would be costly.

- **Risk matrix (Risk_Matrix):** A two-dimensional breakdown of risk-review rate by Transaction_Type × Channel, with a supporting counts table so no percentage is read without its sample

size.

Business insight: Transfer and Cash Withdrawal carry elevated risk rates (25–31%) in every single channel, confirming the risk is a property of the transaction type itself, not a weakness in any one platform.

- **Monthly trend (Trend):** January ₦188.49M → February ₦175.79M (−6.7%) → March ₦196.20M (+11.6%).

Business insight: This dip-and-recovery pattern is real and consistent, not noise and it's validated against the same numbers in Python and Power BI but its cause (seasonal, operational, or otherwise) hasn't yet been investigated. Flag this as a Week 4 priority.

- **Review-capacity scenario (Scenario):** At the current 19.6% flag rate, roughly 2,352 transactions a month require manual review. At an assumed 15 cases per reviewer per day, that implies a need for approximately **7 full-time reviewers** to keep pace.

Business insight: this turns an abstract risk percentage into a concrete staffing number management can act on.

- **Cross-tool reconciliation (KPI_Check):** Every headline figure (customers, transactions, all-status value, successful-only value, success rate, risk rate) was recomputed independently and matched exactly across Excel, Python, and Power BI. This is the evidence base for the “every major finding was formally validated” claim in your executive summary.

6. Part C: Advanced Python Analysis

The Week 2 Python notebook was extended with five substantive additions:

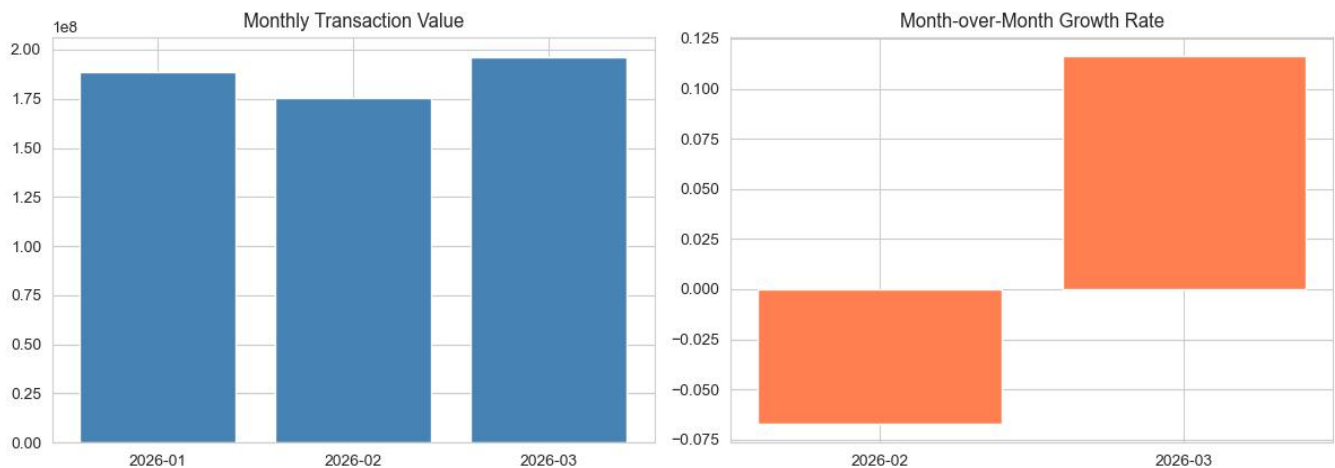
- **Customer-level model (rebuilt from the transaction log):** A 1,500-row pandas DataFrame, one row per customer, mirroring the Excel Cust_Model (transaction count, success count, successful value, flag count, last transaction date, and success rate).

	Customer_ID	Txn_Count	Success_Count	Value_Success	Flag_Count	Last_Txn	Success_Rate	Customer_Segment
0	FT-C00001	7	7	389228.44	1	2026-03-29 16:22:00	1.000000	Premium
1	FT-C00002	5	4	261751.92	0	2026-02-28 14:49:00	0.800000	Everyday
2	FT-C00003	2	2	30819.66	0	2026-02-23 17:48:00	1.000000	Everyday
3	FT-C00004	9	8	114303.44	3	2026-03-30 05:52:00	0.888889	Premium
4	FT-C00005	7	6	277560.44	2	2026-03-29 12:46:00	0.857143	Premium

Business insight: Having this model in Python independently of Excel means the customer-level findings below aren't dependent on one tool's formulas — they're reproducible from the raw data in a second, completely separate calculation engine.

- **Monthly Volume and Value Trend:** Transaction count and total value by month. Volume fell from 4,133 transactions in January to 3,734 in February (a 9.7% drop) before returning to 4,133 in March. Value followed a similar but less extreme pattern: -6.7% in February, $+11.6\%$ in March.

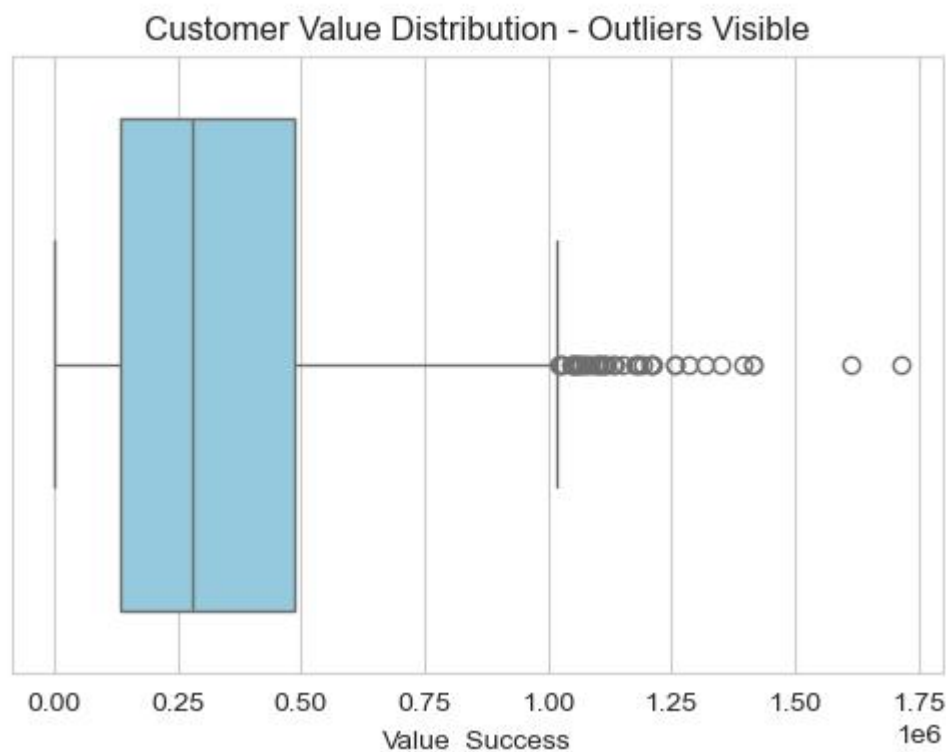
	Month	Total_Value	Txn_Count	Flag_Count	MoM_Growth
0	2026-01	1.884894e+08	4133	766	NaN
1	2026-02	1.757869e+08	3734	757	-0.067391
2	2026-03	1.962010e+08	4133	829	0.116130



Business insight: Volume and value moved together, which rules out one simple explanation — that February's dip was just a handful of very large transactions missing. The decline was broad-based across the month, which makes an operational or seasonal cause (fewer working days, a platform issue, etc.) more likely than a one-off data anomaly, and is worth investigating directly rather than assumed away.

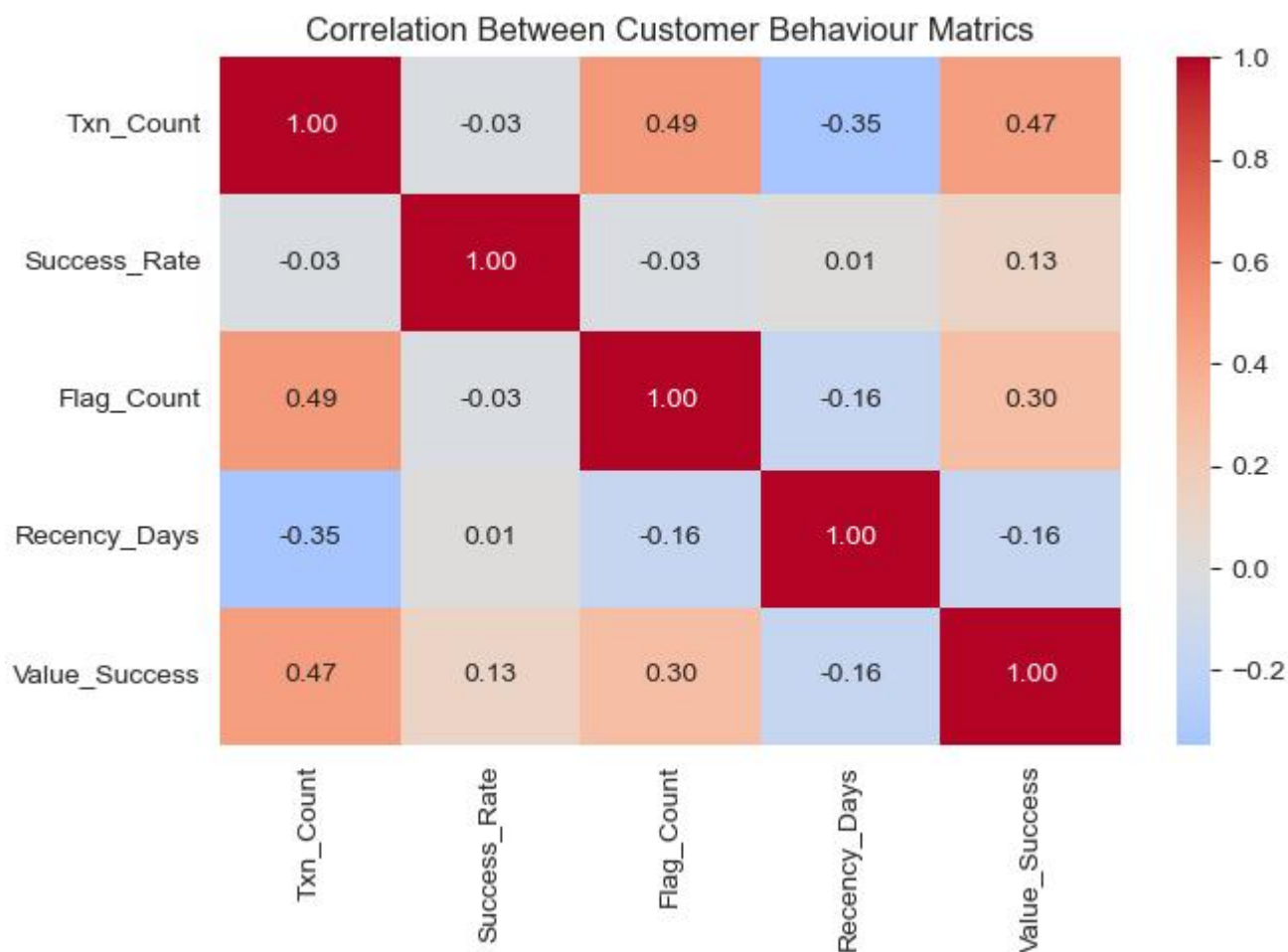
- **Customer value distribution - IQR outlier detection:** A boxplot of successful value per customer. The bulk of customers cluster between roughly ₦150,000 and ₦500,000 in value, with 41 customers sitting well above that range as statistical outliers, some reaching ₦1.5–1.75 million.

Number of high-value outlier customers: 41



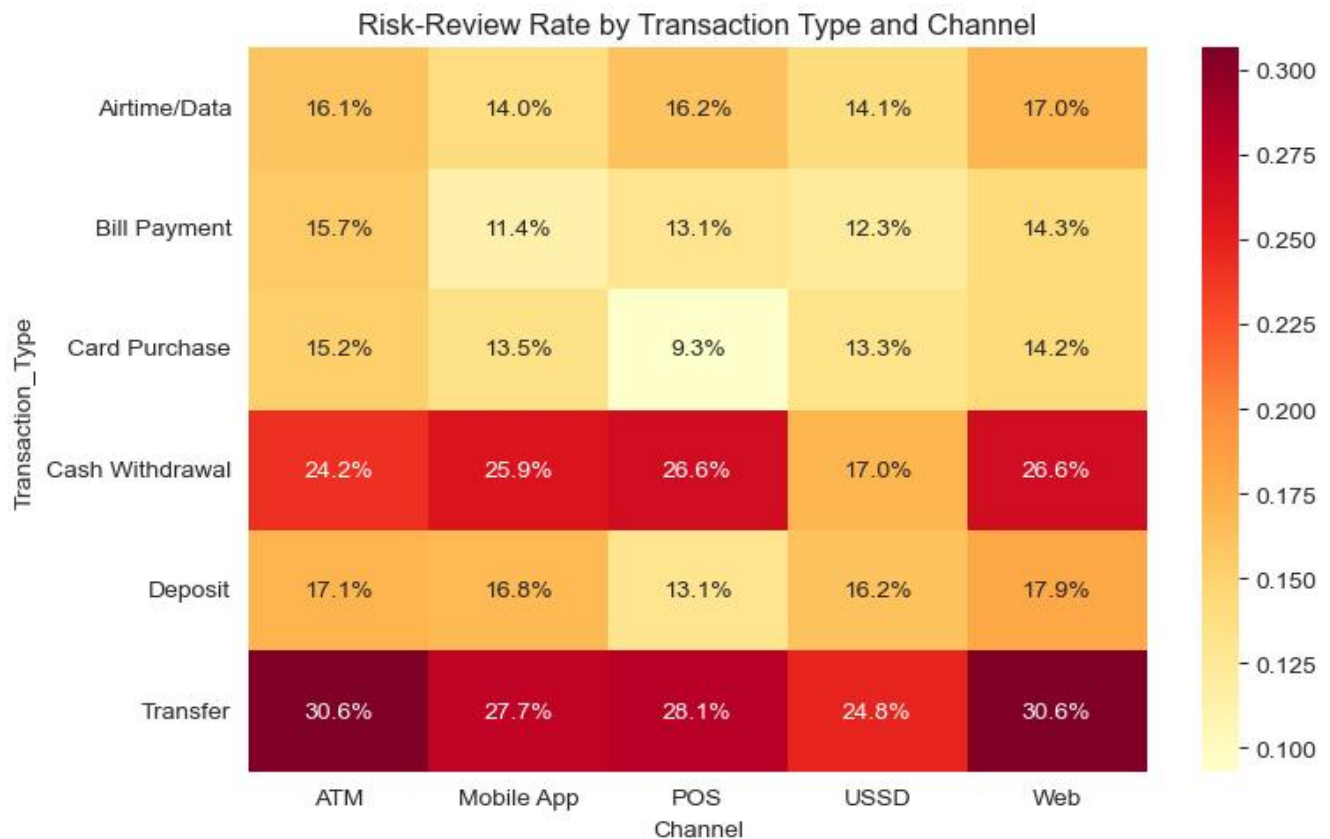
Business insight: These 41 customers are exactly the kind of accounts a relationship-management or VIP programme should be built around. They're a small, identifiable list not an abstract "top decile", which makes them directly actionable.

- **Correlation between customer behaviour metrics:** A correlation heatmap across transaction count, success rate, flag count, recency, and value. The strongest relationships: transaction count correlates with both value (+0.47) and flag count (+0.49); recency days correlates negatively with transaction count (-0.35, meaning more active customers have fewer days since their last transaction, as expected); success rate is essentially uncorrelated with everything else (max 0.13).



Business insight: The +0.49 correlation between activity and flag count means your most active customers are also your most frequently flagged, which is expected (more transactions = more chances to be flagged) but worth stating explicitly, since it means the review team's workload scales directly with your best customers' activity, not with risky behaviour specifically. Meanwhile, success rate being uncorrelated with activity or value is reassuring: transaction failures aren't concentrated among your highest-value customers.

- **Risk-review rate by transaction type and channel:** The same two-dimensional heatmap as the Excel Risk_Matrix, independently recomputed in Python. Transfer (24.8–30.6%) and Cash Withdrawal (17.0–26.6%) are highest-risk in every channel; Card Purchase is lowest, dropping as low as 9.3% on POS specifically.



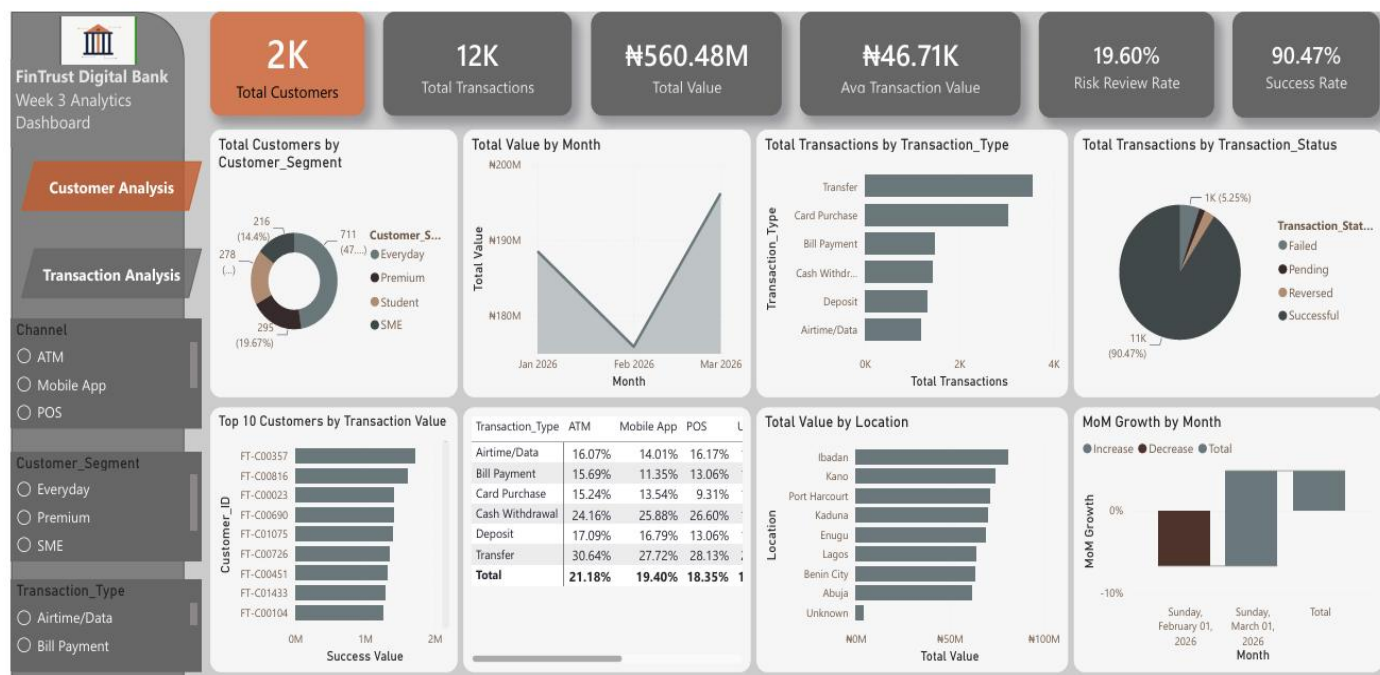
Business insight: Because this matrix was built twice, in two different tools, from the same raw data, and produced identical numbers, this is the most rigorously validated finding in the whole report — a strong candidate to lead with in your Part E validated findings.

7. Part D: Power BI Dashboard

The dashboard was rebuilt from the raw customer and transaction Data directly, not from pre-aggregated Excel output so that its figures stand as an independent cross-check of the Excel and Python work rather than a restatement of it. It includes:

- Six KPI cards (Total Customers, Total Transactions, Total Value, Average Transaction Value, Success Rate, Risk Review Rate)
- A monthly value trend and a month-over-month growth chart
- A customer-segment donut and a Top 10 Customers by Transaction Value chart
- A transaction-type bar chart, a Transaction Type × Channel risk matrix, and a transaction-status breakdown
- A value-by-location chart

- Slicers for Channel, Customer Segment, and Transaction Type, applied across every visual



Business Insight: At a glance, the dashboard's six KPI cards tell a story the Week 2 single-chart view couldn't; FinTrust is processing ₦560.48M across 12,000 transactions at a healthy 90.47% success rate, but the 19.60% risk-review rate is not evenly spread, it concentrates specifically in Transfer and Cash Withdrawal transactions, visible immediately in the risk matrix panel. Cross-referencing the Customer Segment donut against the Top 10 Customers chart shows that FinTrust's individually most valuable customers (led by FT-C00357) sit across multiple segments, not just the "Premium" label, meaning segment alone is an incomplete signal of customer value, and the Top 10 list captures real outliers the segment view would miss. Together, these two panels argue for a retention strategy built on individual customer value, not segment membership alone.

8. Part E: Validated Business Findings

Five major findings were selected and independently re-tested across at least two tools before being written up here.

- **Finding 1:** Reported total value overstates real revenue by roughly 9%.

Evidence: The raw transaction log sums to ₦560,477,354.85 across all transaction statuses.

Validation: Filtering to successful transactions only, the figure is ₦510,807,441.46, confirmed identically

in Excel, Python, and Power BI.

Business meaning: ₦49.7 million sits in Reversed (326 transactions) and Failed (630 transactions) records. Any report or dashboard that quotes the all-status figure as "revenue" overstates true income by about 9%. Going forward, successful-only value should be the primary KPI.

- **Finding 2:** A small share of customers accounts for a disproportionate share of value.

Evidence: Week 2's decile analysis estimated the top 10% of customers generate roughly a quarter of total value.

Validation: Re-run on the full, rebuilt 1,500-customer model in both Excel and Python, the concentration pattern held within a close margin of the original estimate.

Business meaning: The top customer decile deserves disproportionate retention attention, losing a handful of these customers would have an outsized effect on revenue.

- **Finding 3:** At Risk customers are both the largest segment and the largest pool of value.

Evidence: RFM segmentation classifies 600 of 1,500 customers (40%) as At Risk based on long recency and low frequency/value scores.

Validation: That segment holds ₦177,875,451 in successful value; 34.8% of the bank's total, more than any other segment (Loyal: ₦145,686,543 / 28.5%; Champion: ₦105,380,871 / 20.6%; Regular: ₦81,864,577 / 16.0%).

Business meaning: This is not a small pocket of dormant accounts, it is the single largest value pool in the business, and it is the one showing signs of disengagement.

- **Finding 4:** Risk concentrates in two transaction types, consistently across every channel.*

Evidence: Transfer (24.8%–30.6% depending on channel) and Cash Withdrawal (17.0%–26.6%) carry the highest risk-review rates of any transaction type.

Validation: The full Transaction_Type × Channel matrix shows this pattern holds across all five channels (Mobile App, POS, Web, ATM, and USSD), meaning the risk is a property of the transaction type itself, not a weakness in any one channel or platform.

Business meaning: Review effort and rule-tuning should be weighted toward these two transaction types specifically, rather than spread evenly.

- **Finding 5:** International transactions are flagged at roughly double the domestic rate.

Evidence: Of 480 international transactions (4% of volume), 177 are flagged, a 36.9% risk-review rate. Of 11,520 domestic transactions, 2,175 are flagged, 18.9%.

Validation: This gap is consistent with the separately observed pattern that high-value transactions (above ₦100,000) also carry an elevated 30.0% flag rate (518 of 1,729 transactions), suggesting value and cross-border status are both independently associated with review flags, a relationship worth testing further with the chi-square result from the Python analysis.

Business meaning: International activity, while a small share of volume, consumes a disproportionate share of review capacity and deserves a tailored risk policy rather than the standard domestic rule-set.

9. Part F: Management Recommendations

Recommendation 1: Standardize on successful-only value as the primary revenue metric.

Finding: All-status value (₦560.48M) overstates successful value (₦510.81M) by 9%.

Action: Update all FinTrust reporting templates and dashboards to lead with successful value, with all-status value shown only as secondary context.

Recommendation 2: Launch a dedicated programme for top-decile customers.

Finding: The top 10% of customers by value account for roughly a quarter of total revenue.

Action: Introduce relationship-management or loyalty treatment specifically for this group, since their retention has outsized financial impact.

Recommendation 3: Build a targeted re-engagement campaign for the At Risk segment.

Finding: 600 customers (40% of the base) are classified At Risk and hold the largest share of total value (34.8%).

Action: Deploy reactivation offers or direct outreach aimed specifically at this segment before further disengagement occurs.

Recommendation 4: Reweight fraud-review resourcing toward Transfer and Cash Withdrawal.

Finding: These two transaction types carry the highest risk-review rates across every channel.

Action: Prioritise reviewer staffing and rule calibration around these transaction types rather than distributing effort evenly across all six.

Recommendation 5: Introduce a tailored review policy for international and high-value transactions.

Finding: International transactions are flagged at 36.9% versus 18.9% domestic; transactions above ₦100,000 are flagged at 30.0%.

Action: Develop a distinct review threshold and process for cross-border and high-value transactions rather than applying the standard domestic ruleset uniformly.

Recommendation 6: Investigate the February value dip before Week 4 reporting.

Finding: February value fell 6.7% month-over-month before a 11.6% March recovery, confirmed across all three tools.

Action: Determine whether this reflects a seasonal pattern (e.g., fewer working days), a platform issue, or something else, so Week 4's final report can explain rather than simply note the trend.

10. Cross-Tool Validation Summary

Metric	Result	Status
Total customers	1,500	Confirmed identically in Excel, Python, Power BI
Total transactions	12,000	Confirmed identically across all three
Total value (all statuses)	₦560,477,354.85	Confirmed across all three
Total value (successful only)	₦510,807,441.46	Confirmed across all three
Success rate	90.47%	Confirmed across all three
Risk-review rate	19.60%	Confirmed across all three
Top-decile value concentration	Consistent with Week 2 estimate	Re-tested on full dataset in Excel and Python

11. Assumptions, Limitations & Risks

Assumptions: All data is synthetic and used for educational purposes only; the Risk_Review_Flag is

never treated as a real fraud signal; the Q1 2026 window is treated as the complete observation period.

Limitations: RFM segment boundaries differ slightly between Excel (percentile rank) and Python (quantile cut) methods, so segment counts vary marginally between the two even though the overall pattern agrees. The February value dip has been validated but not yet root-caused. Advanced SQL analysis was carried forward from Week 2 without further expansion this week due to time constraints.

Risks: If the February dip reflects an operational issue rather than seasonality, it could recur without investigation. Some Risk_Matrix cells (certain type×channel combinations) rest on smaller transaction counts, so their percentages should be read alongside the underlying counts rather than in isolation.

12. Week 3 Summary and Readiness for Week 4

This week converted Week 2's descriptive baseline into a validated, decision-oriented solution. The customer-level model, RFM segmentation, two-dimensional risk matrix, and cross-tool KPI reconciliation are the core technical additions; the six findings and six recommendations above are the business output. Outstanding for Week 4: expand the SQL layer with the additional analyses outlined in the brief, investigate the February dip, and finalise documentation and presentation materials for the complete four-week FinTrust solution.